English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

Problem 1.

1. (0.25 points) Points A, B are chosen randomly inside a square of side R. Find the probability that the circle centred at A with radius $|AB|$ lies inside the square.

Without loss of generality, $R = 1$. The circle lies in the square if the distance $|AB|$ does not exceed the distance S from A to its boundary. Let $A = (X, Y) \sim \text{Unif}([0,1]^2)$. Then $S = \min\{X, 1-X, Y, 1-Y\}$. For $s \geq 0$,

$$\mathbb{P}(S \leq s) = 1 - \mathbb{P}(S > s) = 1 - \mathbb{P}(s < X < 1-s)\mathbb{P}(s < Y < 1-s)\mathbb{1}_{\{s<1-s\}} = 1 - (1-2s)^2\mathbb{1}_{\{s<1-s\}}.$$

Geometrically, points for which $S < s$ lie in the shaded region:

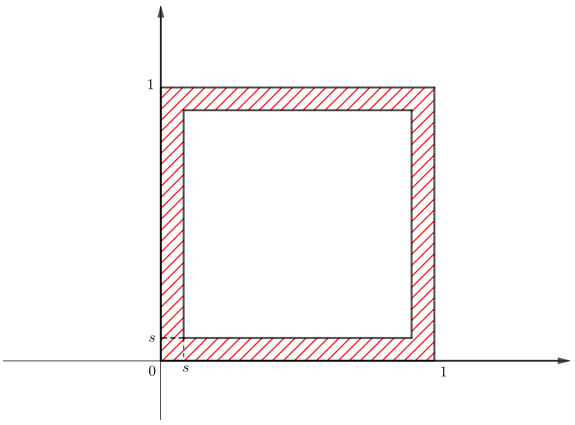


Figure 1: Diagram for **Problem 1.1.**

Thus S is continuous with density $f(s) = 4(1-2s)\mathbb{1}_{(0,1/2)}(s)$. If F denotes the event of interest,

$$\mathbb{P}(F) = \int_{-\infty}^{\infty} \underbrace{\mathbb{P}(F|S=s)}_{\pi s^2} f(s)\, ds = 4\pi \int_0^{1/2} (s^2 - 2s^3)\, ds = \frac{\pi}{24}.$$

2. (0.5 points) Let $X \sim \text{Unif}(0,1)$ and $Y := X(1-X)$. Find $f_Y, F_Y, \mathbb{E}[Y], \text{Var}(Y)$.

The function $g(x) = x(1-x)$ is not monotone, so the change-of-variables formula cannot be applied directly. It is monotone on $[0, 1/2]$ and $[1/2, 1]$. Apply the formula on each interval and add the resulting densities.

On $[0, 1/2]$, the inverse is $h_1(y) = \frac{1}{2} - \frac{1}{2}\sqrt{1-4y}$ for $y \in [0, 1/4]$; on $[1/2, 1]$, it is $h_2(y) = \frac{1}{2} + \frac{1}{2}\sqrt{1-4y}$ on the same range (see below).

For $y \in [0, 1/4]$,

$$f_Y(y) = f_1(y) + f_2(y) = f_X(h_1(y))|h_1'(y)| + f_X(h_2(y))|h_2'(y)|.$$
$$= \mathbb{1}_{(0,1)}(h_1(y)).\frac{1}{\sqrt{1-4y}} + \mathbb{1}_{(0,1)}(h_2(y)).\frac{1}{\sqrt{1-4y}}.$$

It remains to determine where the indicators equal 1. The equation $\mathbb{1}_{(0,1)}(h_i(y)) = 1$, $i \in \{1, 2\}$, is equivalent to $0 < 1 + (-1)^i\sqrt{1-4y} < 2$, which holds whenever $y \in (0, 1/4)$. Hence

$$f_Y(y) = \frac{2}{\sqrt{1-4y}}\mathbb{1}_{(0,1/4)}(y).$$

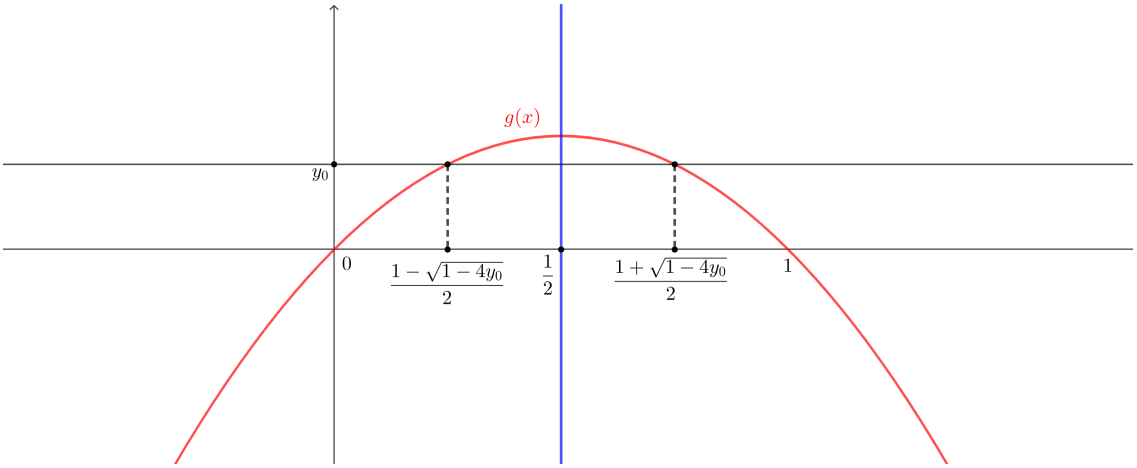


Figure 2: Diagram for **Problem 1.2.**

Alternatively, for $y \in [0, 1/4]$,

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\left(X(1-X) \le y, X \le 1/2\right) + \mathbb{P}\left(X(1-X) \le y, X > 1/2\right)$$

$$= \mathbb{P}\left(X \le \frac{1-\sqrt{1-4y}}{2}\right) + \mathbb{P}\left(X > \frac{1+\sqrt{1-4y}}{2}\right) = \frac{1-\sqrt{1-4y}}{2} + 1 - \frac{1+\sqrt{1-4y}}{2} = 1 - \sqrt{1-4y}$$

For $y < 0$, $F_Y(y) = 0$, and for $y > 1/4$, $F_Y(y) = 1$. We can also obtain $f_Y = F_Y'$ almost everywhere this way. Furthermore, $\mathbb{E}Y = \mathbb{E}X - \mathbb{E}X^2 = 1/6$, $\mathbb{E}Y^2 = \mathbb{E}[X^2(1-X)^2] = \mathbb{E}[X^2 - 2X^3 + X^4] = 1/30$, and $\mathrm{Var}(Y) = 1/180$.

3. (0.5 points) Let X, Y have joint density

$$f_{XY}(x,y) := \begin{cases} c(x+y), & x \ge 0, y \ge 0, x+y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Define the random variables

$$U := X + Y, \qquad V := 1 - \frac{Y}{U}.$$

Find $f_{U,V}$, $\mathrm{Cov}(U,V)$, $\mathbb{E}[X|U]$ and $\mathbb{E}[U|Y]$.

For c,

$$1 = \iint f_{X,Y}(x,y)\,dx\,dy = \int_0^1 \int_0^{1-x} c(x+y)\,dy\,dx = c\int_0^1 \left(x - x^2 + \frac{(1-x)^2}{2}\right) dx = \frac{c}{3}.$$

Thus $c = 3$. Consider $(x,y) \mapsto (x+y, 1 - y/(x+y)) = (x+y, x/(x+y))$, whose inverse is $(u,v) \mapsto (uv, u - uv)$. Its Jacobian is

$$J = \begin{vmatrix} v & u \\ 1-v & -u \end{vmatrix} = -u \ne 0 \quad \text{for } x, y \in D_{X,Y} \setminus \{(0,0)\}.$$

The transformation formula gives

$$f_{U,V}(u,v) = f_{X,Y}(uv, u-uv).|J| = 3(uv + u - uv).|u|.\mathbb{1}_{\{0 \le uv \le 1,\ 0 \le u - uv \le 1 - uv\}}.$$

The support requires $0 \le uv \le 1$ and $uv \le u \le 1$, i.e. $0 \le uv \le u \le 1$. Thus $0 \le u, v \le 1$ (which automatically implies $uv \le u$). Hence $f_{U,V}(u,v) = 3u^2\mathbb{1}_{(0,1)^2}(u,v)$. The density does not depend on v except through its support. Its marginal densities are $f_V(v) = \mathbb{1}_{(0,1)}(v)$, so $V \sim \mathrm{Unif}(0,1)$, and $f_U(u) = 3u^2\mathbb{1}_{(0,1)}(u)$. Hence $U \perp\!\!\!\perp V$ because $f_{U,V} = f_U f_V$, and $\mathrm{Cov}(U,V) = 0$.

For the conditional expectations, use conditional densities. The transformation $(x,y) \mapsto (x, x+y)$ gives

$$f_{X,U}(x,u) = f_{X,Y}(x, u-x)\begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 3(x+u-x)\mathbb{1}_{(0,1)}(u-x)\mathbb{1}_{(0,1-(u-x))}(x) = 3u\mathbb{1}_{(0,1)}(u)\mathbb{1}_{(0,u)}(x).$$

Since $f_U(u) = 3u^2\mathbb{1}_{(0,1)}(u)$, for $u \in (0,1)$,

$$f_{X|U}(x|u) = \frac{3u\mathbb{1}_{(0,u)}(x)}{3u^2} = u^{-1}\mathbb{1}_{(0,u)}(x).$$

Now

$$\mathbb{E}[X|U=u] = \int_{-\infty}^{\infty} x f_{X|U}(x|u)\,dx = \int_0^u \frac{x}{u}\,dx = \frac{u}{2}$$

for $u \in (0,1)$. Thus $\mathbb{E}[X|U] = U/2$ (see below). Alternatively, $\mathbb{E}[X|U] = \mathbb{E}[UV|U] = U\mathbb{E}[V] = U/2$. For the second expectation, $\mathbb{E}[X+Y|Y] = \mathbb{E}[X|Y] + \mathbb{E}[Y|Y] = \mathbb{E}[X|Y] + Y$, so it suffices to calculate $\mathbb{E}[X|Y]$.

$$f_Y(y) = \int_0^{1-y} 3(x+y)dx \; .\mathbb{1}_{(0,1)}(y) = \frac{3}{2}\left(1-y^2\right).\mathbb{1}_{(0,1)}(y),$$

$$\Rightarrow f_{X|Y}(x|y) = \begin{cases} \frac{3(x+y)}{3/2(1-y^2)} \cdot \mathbb{1}_{(0,1-y)}(x), & y \in (0,1) \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\mathbb{E}[X|Y=y] = \int_0^{1-y} x\frac{2x+2y}{1-y^2}\,dx = \frac{2}{3(1-y^2)}x^3\Big|_0^{1-y} + \frac{y}{1-y^2}x^2\Big|_0^{1-y} = \frac{2-y-y^2}{3+3y},$$

so

$$E[U|Y] = \frac{2-Y-Y^2}{3+3Y} + Y = \frac{2}{3}\frac{1+Y+Y^2}{1+Y}.$$

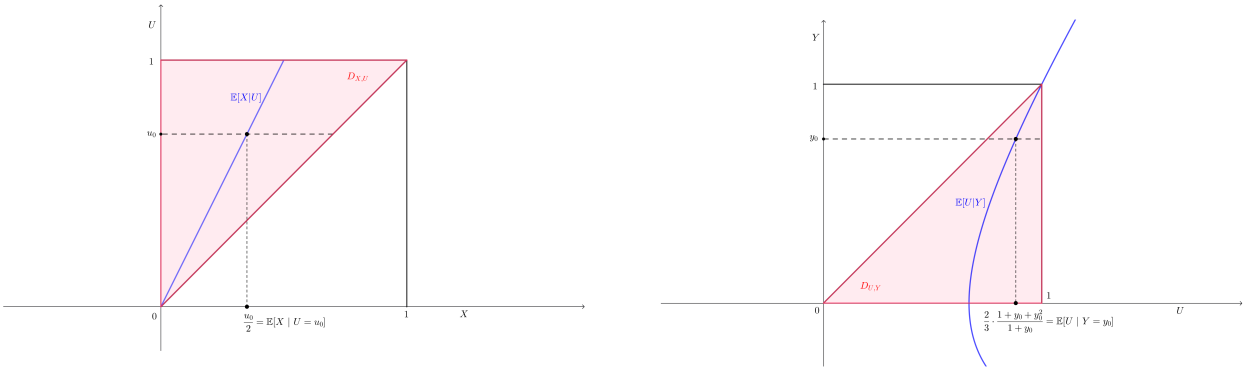


Figure 3: The conditional expectations $\mathbb{E}[X|U]$ and $\mathbb{E}[U|Y]$ are shown in blue.

Problem 2. A metro station has an electronic timetable display with 48×40 LEDs. Their lifetimes are independent exponential variables with mean 40,000 operating hours. The display operates 18 hours daily. If fewer than 85% of the LEDs work, the text becomes hard to read and the display must be replaced.

1. (0.25 points) Approximate the probability that the display needs replacement after 365 days.

Let X_i be the lifetime in hours of LED i, $1 \le i \le 48 \times 40$. The X_i are iid Exp(1/40 000). The number working at time t is

$$T_t = \sum_{i=1}^{48\times 40} \mathbb{1}_{\{X_i \ge t\}} \sim \text{Bin}(48 \times 40, \mathbb{P}(X_i \ge t)).$$

Replacement after 365 days at 18 hours daily means $T_{365\times 18} < 85\% \times 48 \times 40 = 1632$. Here $p = \mathbb{P}(X_i \ge 365 \times 18) = \exp(-365 \times 18/40\,000)$. Thus $\mu = \mathbb{E}[T_{365\times 18}] = 48 \times 40 \exp(-365 \times 18/40\,000) \approx 1629.18$ and $\sigma = \sqrt{\text{Var}(T_{365\times 18})} = \sqrt{\mu(1-p)} \approx 15.71$. The CLT gives the approximation

$$\mathbb{P}(T_{365\times 18} < 85\% \times 48 \times 40) \approx \mathbb{P}\left(\frac{T_{365\times 18} - \mu}{\sigma} < \frac{1632 - 1629.18}{15.71}\right) \approx \Phi(0.1795) \approx 57.142\%.$$

2. (0.5 points) Let T_{min} be the time in operating hours to the first LED failure. Find its distribution and $\mathbb{E}[T_{min}]$, the probability of the first failure within k calendar days, and its median.

We have $T_{min} = \min\{X_i : 1 \le i \le 48 \times 40\}$. For independent $X \sim \text{Exp}(\alpha)$ and $Y \sim \text{Exp}(\beta)$, $\min\{X,Y\} \sim \text{Exp}(\alpha+\beta)$, since $\mathbb{P}(\min\{X,Y\} > t) = \mathbb{P}(X > t)\,\mathbb{P}(Y > t) = e^{-\alpha t}e^{-\beta t} = e^{-(\alpha+\beta)t}$. Together with $\min\{X_i : 1 \le i \le n\} = \min\{X_n, \min\{X_i : 1 \le i \le n-1\}\}$, this yields $T_{min} \sim \text{Exp}(48 \times 40/40\,000)$ and $\mathbb{E}[T_{min}] = 40\,000/(48 \times 40) \approx 20.83$.

$$\mathbb{P}(T_{min} \le 18k) = 1 - e^{-48\times 40\times 18k/40\,000}.$$

For an exponential variable of rate $\lambda > 0$, the median m satisfies

$$\frac{1}{2} = \mathbb{P}(X \le m) = 1 - e^{-\lambda m} \quad \Rightarrow \quad -\lambda m = -\ln 2 \quad \Rightarrow \quad m = \frac{\ln 2}{\lambda}.$$

In particular, the median of T_{min} is $(40\,000 \ln 2)/(48 \times 40) \approx 14.44$.

3. (0.5 points) After 12 months, exactly 1700 LEDs are found working. What is the distribution of the remaining lifetime of an individual LED at this time? Find the probability that the display becomes unreadable within the next three months.

It is natural to ask about the lifetime of LED i if it *has not failed* at time t (here $t = 365 \times 18$). For $s > 0$,

$$\mathbb{P}\left(X_i > t+s | T_t = 1700\right) = \frac{\mathbb{P}\left(X_i > t+s, T_t = 1700\right)}{\mathbb{P}\left(T_t = 1700\right)} = \frac{\mathbb{P}\left(X_i > t+s, \sum_{j\neq i} \mathbb{1}_{\{X_j \geq t\}} = 1699\right)}{\mathbb{P}\left(T_t = 1700\right)},$$

because $X_i > t+s$ implies $\mathbb{1}_{\{X_i \geq t\}} = 1$. Mutual independence allows the numerator to factor:

$$\mathbb{P}\left(X_i > t+s | T_t = 1700\right) = \frac{\mathbb{P}\left(X_i > t+s\right)\mathbb{P}\left(T_t^{(i)} = 1699\right)}{\mathbb{P}\left(T_t = 1700\right)},$$

where $T_t^{(i)} = T_t - \mathbb{1}_{\{X_i \geq t\}}$, so $T_t^{(i)} \sim \text{Bin}(n-1, p)$[1]. Calculate:

$$\mathbb{P}\left(X_i > t+s \mid T_t = 1700\right) = \frac{e^{-(t+s)/40\ 000}\binom{n}{1699}p^{1699}(1-p)^{n-1699}}{\binom{n}{1700}p^{1700}(1-p)^{n-1700}}$$

$$= \frac{1700}{n-1699} \cdot \frac{1-p}{p} \cdot e^{-(t+s)/40\ 000} = \frac{1700}{221}\left(e^{-s/40\ 000} - e^{-(t+s)/40\ 000}\right),$$

The density conditional on $\{T_t = 1700\}$ can also be obtained from this.

One could also ask about the conditional distribution of the time $\widetilde{X}_i$ to failure for an LED that *has already failed* at t. For $0 < s < t$,

$$\mathbb{P}\left(\widetilde{X}_i < -s | T_t = 1700\right) = \mathbb{P}\left(X_i < t-s | T_t = 1700\right) = \frac{\mathbb{P}\left(X_i < t-s\right)\mathbb{P}\left(T_t^{(i)} = 1700\right)}{\mathbb{P}\left(T_t = 1700\right)},$$

again using independence and the fact that $X_i < t-s$ implies $\mathbb{1}_{\{X_i \geq t\}} = 0$. This can be calculated similarly.

For the final question, calculate $\mathbb{P}\left(T_{t+90\times 18} < 1632 | T_t = 1700\right)$, where $t = 365 \times 18$. More than 68 LEDs must fail in those three months (so $t \leq X_i < t + 90 \times 18$ for them). Writing $F(t) = \mathbb{P}\left(X_1 > t | T_t = 1700\right)$ as calculated above, the required probability is

$$\sum_{k=69}^{1700} \binom{1700}{k} \left(1 - F(t + 90\times 18)\right)^k \left(F(t+90\times 18)\right)^{1700-k}.$$

[1] Here $p = \mathbb{P}\left(X_1 \geq t\right) = e^{-t/40\,000}$ and $n = 48 \times 40$.

Problem 3. At a juice-bottling plant, the volume X in one bottle (millilitres) is normally distributed with variance $\sigma^2 = 12^2$.

1. (0.25 points) If 20% of bottles contain less than 485 ml, find μ. Find the percentage containing more than 525 ml.

$$0.2 = \mathbb{P}(X < 485) = \Phi\left(\frac{485-\mu}{12}\right) \;\Rightarrow\; \frac{485-\mu}{12} \approx -0.84 \;\Rightarrow\; \mu \approx 495.08.$$

$$\mathbb{P}(X > 525) = 1 - \mathbb{P}(X \le 525) = 1 - \Phi\left(\frac{525-495.08}{12}\right) \approx 1 - \Phi(2.493) = \Phi(-2.493) \approx 0.64\%.$$

2. (0.25 points) After a production change, assume $X \sim \mathrm{N}(505, \sigma^2)$. Find σ if 95% of bottles contain between 490 and 520 ml.

$$0.95 = \mathbb{P}(490 < X < 520) = \mathbb{P}\left(-\frac{15}{\sigma} < \frac{X-505}{\sigma} < \frac{15}{\sigma}\right) = \Phi\left(\frac{15}{\sigma}\right) - \Phi\left(-\frac{15}{\sigma}\right) = 1 - 2\Phi\left(-\frac{15}{\sigma}\right),$$

$$\Rightarrow \Phi\left(-\frac{15}{\sigma}\right) = 0.025 \;\Rightarrow\; -\frac{15}{\sigma} \approx -1.96 \;\Rightarrow\; \sigma \approx 7.65.$$

3. (0.5 points) After the change, with σ from the preceding part, use two-stage quality control. First, randomly choose $n = 30$ bottles from a batch and measure their volumes. Reject the batch if at least three contain under 490 ml. Otherwise, draw another $m = 20$ bottles and calculate the mean volume of all $m + n = 50$. Reject if this mean is below 503 ml, and accept otherwise. Find the rejection probability.

Let X_i, $1 \le i \le 50$, be the volume in bottle i. As in Problem 2, put

$$T = \sum_{i=1}^{30} \mathbb{1}_{\{X_i < 490\}} \sim \mathrm{Bin}(30, p),$$

where $p = \mathbb{P}(X < 490) = \Phi(-15/\sigma) = 0.025$. Let A be the event of rejecting the batch. Then

$$\mathbb{P}(A) = \mathbb{P}(T \ge 3) + \mathbb{P}\left(T < 3, \overline{X}_{50} < 503\right) = 1 - \mathbb{P}\left(T < 3, \overline{X}_{50} \ge 503\right).$$

Problem 4. (0.75 points) A server's load Y is exponentially distributed with rate $\lambda > 0$. Given $Y = y$, a task's processing time X (seconds) is exponential with rate y. Find f_X, $\mathbb{P}(X > x)$, $\mathbb{E}[X]$ and the distribution of $X + Y$. Is $X + Y$ a continuous random variable?

By assumption, $X|Y = y \sim \mathrm{Exp}(y)$, so $f_{X|Y}(x|y) = ye^{-yx}\mathbb{1}_{(0,\infty)}(x)$. Hence

$$f_{X,Y}(x,y) = f_Y(y) f_{X|Y}(x|y) = \lambda e^{-\lambda y} y e^{-yx} \mathbb{1}_{(0,\infty)}(x)\mathbb{1}_{(0,\infty)}(y) = \lambda y e^{-(\lambda+x)y} \mathbb{1}_{(0,\infty)^2}(x,y),$$

and the marginal density of X for $x > 0$ is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy = \int_0^{\infty} \lambda y e^{-(\lambda+x)y}\,dy = \underbrace{\int_0^{\infty} y(\lambda+x)e^{-(\lambda+x)y}\,dy}_{(\lambda+x)^{-1}} - x\int_0^{\infty} ye^{-(\lambda+x)y}\,dy =$$

[2]

$$= \frac{1}{\lambda+x} + \underbrace{\frac{x}{\lambda+x} y e^{-(\lambda+x)y}\Big|_{y=0}^{\infty}}_{0} - \frac{x}{\lambda+x}\int_0^{\infty} e^{-(\lambda+x)y}dy = \frac{1}{\lambda+x} + \frac{x}{(\lambda+x)^2} e^{-(\lambda+x)y}\Big|_{y=0}^{\infty} = \frac{\lambda}{(\lambda+x)^2}.$$

Thus, for $x > 0$,

$$\mathbb{P}(X > x) = \int_x^{\infty} \frac{\lambda}{(\lambda+t)^2}\,dt = \frac{\lambda}{\lambda+t}\Big|_{\infty}^{x} = \frac{\lambda}{\lambda+x}, \qquad \mathbb{E}X = \int_0^{\infty} \frac{\lambda x}{(\lambda+x)^2}\,dx = +\infty,$$

because the last integral diverges. For the distribution of $X + Y$, when $s > 0$,

$$\mathbb{P}(X+Y \le s) = \int_{-\infty}^{\infty} \mathbb{P}(X+Y \le s | Y = y) f_Y(y)\,dy$$
$$= \int_0^{\infty} \mathbb{P}(X+y \le s | Y = y) \lambda e^{-\lambda y}\,dy$$

[2] The first integral is the expectation of an exponential random variable with rate $\lambda + x$.

$$= \int_0^s (1 - e^{-y(s-y)})\lambda e^{-\lambda y}\, dy.$$

The last equality uses $\mathbb{P}(X + y \leq s | Y = y) = 0$ when $y > s$, and the exponential conditional law, giving $\mathbb{P}(X \leq s - y | Y = y) = 1 - e^{-y(s-y)}$. The CDF is differentiable for $s > 0$, zero for $s \leq 0$, and continuous at 0, hence absolutely continuous. Thus $X + Y$ is continuous with density

$$f_{X+Y}(s) = \frac{d}{ds} F_{X+Y}(s) = \frac{d}{ds}\left(\int_0^s (1 - e^{-y(s-y)})\lambda e^{-\lambda y}\, dy\right) \mathbb{1}_{(0,\infty)}(s) =$$

$$\left(\left(1 - e^{-y(s-y)}\right)\lambda e^{-\lambda y}\Big|_{y=s} + \int_0^s \frac{\partial}{\partial s}\left(1 - e^{-y(s-y)}\right)\lambda e^{-\lambda y} dy\right)\ \mathbb{1}_{(0,\infty)}(s) = \int_0^s y e^{-y(s-y)}\lambda e^{-\lambda y} dy.\ \mathbb{1}_{(0,\infty)}(s).$$